Operator Expressions

Expressions may be combined using operators. The Ruby operators in precedence order are listed in Table 13, Ruby operators (high to low precedence), on page 455. The operators with a ✓ in the Method column are implemented as methods and may be overridden.

Method	Operator	Description
✓	! ~ + -	Not, complement, unary plus (method name for unary plus is +@)
✓	**	Exponentiation
✓	-	Unary minus (method names is -@)
✓	* / %	Multiply, divide, and modulo
✓	+ -	Plus and minus
✓	>> <<	Right and left shift (<< is also the append operator)
✓	&	"And" (bitwise for integers)
✓	^ |	Exclusive "or" and regular "or" (bitwise for integers)
✓	<= < > >=	Comparison operators
✓	<=> == === != =~ !~	Equality and pattern match operators
	&&	Logical "and"
	||	Logical "or"
	.. ...	Range (inclusive and exclusive)
	? :	Ternary if-then-else
	rescue	When used as a modifier at the end of a line
	= %= /= -= += |= &= >>= <<= *= &&= ||= **= ^=	Assignment
	defined?	Test for if a value is defined in current binding
	not	Logical negation
	or and	Logical composition
	if unless while until	Expression modifiers at the end of line
	{ }	Block expression
	do end	Block expression

Table 13—Ruby operators (high to low precedence)

More on Assignment

The assignment operator assigns one or more *rvalues* (the *r* stands for "right," because rvalues tend to appear on the right side of assignments) to one or more *lvalues* ("left" values). What is meant by assignment depends on each individual lvalue.

As the following shows, if an lvalue is a variable or constant name, that variable or constant receives a reference to the corresponding rvalue.

```
a = /regexp/
b, c, d = 1, "cat", [ 3, 4, 5 ]
```

If the lvalue is an object attribute, the corresponding attribute-setting method will be called in the receiver, passing as a parameter the rvalue: